I-FDP PROPOSAL DOCUMENT	
Group ID: S2_01	**Description**
Group Theme: *Laggy Piggies*	We will implement Laggy Piggies, inspired by Bad Piggies, an interactive vehicle construction and physics puzzle game by Rovio Entertainment. Players will construct a vehicle in a 3×3 grid from a limited inventory of frames, wheels, balloons, a pig and engines, then attempt to reach a goal on a predefined map. Movement will depend on its surviving components, with weight, gravity, balloon lift and terrain collisions influencing its path. TNT can destroy individual parts within its blast radius. Destruction of the pig will end the attempt. Players will interact through the mouse, buttons and switches on the Basys3 board, with instructions explaining the objective and controls. The right OLED will display the map and the vehicle's behaviour. The left OLED will show inventory and the construction grid. During gameplay, it will show component status and allow toggling of the engine and popping of balloons. Short audio effects will accompany key events such as collisions or pig death. Players can retry a level with their original build or return to construction to improve their design or refer to hints. We plan to include two maps.
Group Member 1: *Kaushal Rajesh* Student No.: A0315560L	Member responsibility: Vehicle construction, user input, damage state and audio effects. **Week 7:** Looked through the project document and formed the FDP group. Completed GPT-5 and familiarised with the OLED module. Started a brainstorm of ideas for FDP. **Week 8:** Finalised idea and formalised role division for FDP. Subsequently, selected preferred role. Created OLED manipulation modules, as well as a button debouncer and custom clock divider. Collected a Pmod amplifier and speaker from the lab to prepare for audio implementation. **Week 9:** Cloned the shared GitHub repository to collaborate on code and contributed to the proposal document. Familiarised myself with the mouse module and setup. Implemented a modular code structure to support code sharing and team integration, and a construction grid for placing vehicle parts. **Week 10:** ● Implement inventory accounting and build validation, testing rejected placements and refunds ● Implement damage state during runtime and preserve the original build for retry ● Connect construction data and vehicle properties to teammates' display, movement and game control modules **Week 11:** ● Integrate mouse and button inputs for construction, instructions, toggling the engine and popping balloons ● Implement game sounds with priority handling, prevention of repeated playback and mute/reset controls ● Connect damage handling to teammates' collision, movement and display modules **Week 12:** ● Fix integration issues and stress test both levels on the Basys3 ● Prepare the report and rehearse project demonstration
Group Member 2: *Kyle Lee Yao Yu* Student No.: A0322756B	Member responsibility: Implement the physics engine, vehicle movement, engine acceleration, boost, balloon lift and wheel physics. **Week 7:** ● Formed FDP group and read project document, familiarised with the PMOD OLED module, completed training tasks for GPT 5 **Week 8:** ● Brainstormed ideas for potential project, researched was on how to implement physics engine **Week 9:** ● Create shared github repository to help with collaboration, filled in project proposal ● Implement game object gravity and max fall speed ● Implement acceleration on main vehicle body and parts using a car engine **Week 10:** ● Create speed limit for main vehicle and parts ● Implement wheel movement and wheel drag physics ● Begin integrating movement with Member 3's collision module **Week 11:** ● Implement new movement mechanics, boost and balloon lift

	• Coordinate with Member 3 on combining movement and collisions with the new movement mechanics • Coordinate with Member 1 on integrating buttons with various movement types • Work out integration on movement animation with Member 4 **Week 12:** • Work on starting and restarting vehicle position and movement for level retry • Fix bugs to ensure smooth integration with the other members
Group Member 3: *Lam Yap Ting* Student No.: A0320434U	Member responsibility: Implement collision detection, TNT destruction mechanics and game outcome checks. **Week 7:** Formed FDP group and looked through the project document **Week 8:** Brainstormed project ideas, divided the project into four parts and completed GPT-5 practice tasks on implementing different OLED display behaviours **Week 9:** • Completed my part of the project proposal document, joined and cloned the shared GitHub repository • Researched Axis-Aligned Bounding Box collision detection for vehicle-terrain interactions **Week 10:** • Implement basic collision detection and response using test geometry, like platforms and walls • Test landing detection, boundary collisions and vehicle movement at different speeds, establish the position and velocity inputs for the collision module and begin integration with Member 2 **Week 11:** • Implement TNT triggering, explosion-radius calculations and damage-mask generation • Develop game outcome checks for reaching the goal, hitting hazards and going out of bounds • Coordinate with Member 1 on damage handling for TNTs, Member 2 on vehicle collision integration, and Member 4 on game outcome signals **Week 12:** • Fix remaining integration issues with the other members' modules and test edge cases, simultaneous TNT explosions and retry/reset behaviour • Debug integration issues and verify that the game works reliably for final demonstration
Group Member 4: *Marcus Lim* Student No.: A0271977L	Member responsibility: Graphics, game control and levels: rendering OLEDs, running the game state machine and designing the level maps. **Week 7:** Formed a group of four and set up a group chat to coordinate meetings and work. **Week 8:** Brainstormed game ideas and agreed on a Bad Piggies style game. Split the work into four roles and decide which roles to take. **Week 9:** Joined the team's GitHub repository with a shared folder structure so everyone can work on their own modules. I researched how to build my part in Verilog, including drawing sprites from the OLED pixel index and structuring the game as a state machine. **Week 10:** Build the core of my role: • a game controller for the build, run, success and fail stages • level data for 2 maps with terrain, spawn point, goal and TNT positions • Renderers showing the construction grid and inventory on the left OLED, and the map and vehicle on the right OLED • Begin connecting teammates' construction data and vehicle position to the renderers and game controller **Week 11:** Add instructions, hints and component-status/control graphics on the left OLED, and test both maps. **Week 12:** • Integrate my modules with my teammates' input, movement and collision modules. Run initial tests of the full game loop: build a vehicle, launch it, reach the goal or fail, and retry • Debug issues found during testing and make improvements so the game runs smoothly and reliably before the final demonstration